

Blue Planet Rescue VR: Unite for Marine Preservation

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Figure 1: User Interface in the Blue Planet Rescue VR game.

ABSTRACT

Marine conservation is increasingly urgent due to pollution, over-fishing, and climate change, which pose significant threats to marine ecosystems. Plastic pollution, in particular, has severe negative effects on the survival of sea animals. This paper presents Blue Planet Rescue VR, an immersive virtual reality (VR) game aimed at raising public awareness and prompting action in the face of an environmental crisis. This two-player collaborative game combines interactive mechanics, a heads-up display (HUD), and educational elements to address these pressing issues. Players explore the underwater environment, learning about the harmful effects of pollution on marine ecosystems and are motivated to engage in real-life conservation efforts. The game underscores the necessity of collective action for marine preservation, advocating that unity is essential to safeguarding our blue planet.

Index Terms: Human-Computer Interaction, Virtual Reality, Multi-player game, Marine Conservation.

1 INTRODUCTION

Among Environmental Sustainability themes such as recycling resources, sustainable energy, climate change, and green lifestyles. The immersive VR game has been proven to help the effective implementation of sustainability education [4, 5]. For example, to educate people about a sustainable lifestyle, Louis et al. developed a serious VR game, “Sustainable Futures”, which is designed to enhance awareness and understanding of sustainability challenges

through a thematic educational gaming experience [6]. As one of the UN’s Sustainable Development Goals (SDGs), ocean conservation is intrinsically linked to human survival yet often overlooked by the public. Humanity heavily depends on the ocean’s vital services, including oxygen production and provision of food. However, few people understand how their daily actions impact ocean health. Therefore, our group has chosen to focus on a less addressed but crucial topic, which is Marine Conservation.

After reviewing the previous work, we found that several VR games exist in underwater environments, such as simulating diving experiences [3]. Utilising the immersive feature of VR, Josep et al. created a fully immersive underwater virtual tour using 360-degree images [2]. Although there is a VR application developed by Paolo et al. that addressed the climate change impacts on coral reef inhabitants, the game design is relatively simple, and they haven’t created a collaborative virtual environment [1].

However, marine conservation cannot achieve its ultimate goal without the cooperation of all stakeholders. Marine pollution, in particular, is a serious problem within the scope of conservation. It is primarily caused by littering and intentional discharge of pollutants, which significantly endangers marine organisms and threatens long-term ocean sustainability. Through this project (Fig. 1), our group uses collaborative VR technology to demonstrate the importance of collective behaviours in marine conservation by educating players about the harmful effects of ocean littering and encouraging a plastic-free lifestyle.

2 PROJECT OVERVIEW

The current work was designed in the Unity 6 engine by making use of the VR multiplayer template, which provides a convenient and efficient way to build immersive virtual reality multiplayer experiences. During the development process, the Oculus Quest link 3 was utilized.

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Figure 2: Colour settings for the animal status.

Blue Planet Rescue VR is a multiplayer game in which players collaborate to clean up ocean litter and rescue endangered marine species. The game also features underwater terrain that showcases ecosystems and animals in the sea. Input actions are handled on the left or right controllers, including select, grab, move around, rotation, and teleportation, the same as the presets on the Unity XR Interaction Toolkit.

In the lobby, the game functions as an underwater exhibit, players can move freely and observe the terrain of the underwater world and learn about the elements under the sea by examining sea animals and coral reefs, as well as marine ecology.

Upon game start, two types of objects are spawned in the field: trash and animals; these objects exhibit random movement with different interactable methods. Trash objects are clickable and grabbable items that the player can pick up, while animal objects require interaction by touching the animal object with the hand controller. The colour of the animal object changes (shown in Fig. 2), indicating the stages of animal rescue: blue when it is interactable, yellow when it is rescuing, and green when the rescue is complete. The animal also ascends and disappears after the rescue is complete.

Drawing inspiration from the ocean exploration game *Subnautica*, we implemented a heads-up display (HUD) that presents scores, target counts, and other game metrics directly in the player's field of view, replacing traditional world-space panels and canvases. This design choice significantly enhances immersion by minimizing visual obstruction while maintaining clear communication of game status to players.

3 COLLABORATIVE INTERACTION TECHNIQUES

Our game features collaborative interaction between players by assigning players two roles: trash cleaner and animal rescuer. One is in charge of collecting trash, and another is in charge of rescuing endangered marine organisms. These two players undertake different tasks respectively and collaborate with each other to achieve the game objective. Both of these actions will earn corresponding points for the player team, and the two players share the same score pool. This design aims to promote cooperation among players, enabling them to work towards a common goal, that is, to obtain the highest possible score within the limit oxygen.

Moreover, the number of animals will dynamically decrease as the amount of garbage increases. This is designed to reflect the ecological impact of excessive litter on the marine environment, where the accumulation of litter possible exert detrimental impacts on the living conditions of animals, leading to a reduction in their population. Such a mechanism not only adds a layer of realism to the game but also introduces an additional strategic element, as players need to consider the trade-off between collecting more garbage for points and preserving the animal population to ensure the long-term sustainability of their scoring opportunities. It enriches the game

dynamics by creating a more complex and interconnected gameplay loop, where the actions of one player have implications for the other's gameplay experience, thereby enhancing the overall cooperative nature of the game.

4 DISCUSSION AND CONCLUSION

In short, this project showcases the potential of multiplayer VR technology in environmental education through an innovative and engaging ocean conservation game. The implementation combines networked player interactions, physics-based gameplay, and an immersive HUD system to create an engaging learning experience. It emphasizes the importance of collective efforts.

In the future, we plan to expand the game with additional stages addressing more types of severe marine problem such as managing oil spills or addressing algal blooms. To enhance the educational value and gameplay experience, we will implement interactive features such as informative pop-up texts about marine species and their characteristics. The player HUD will be expanded to include health indicators, gaze-based target controls, and visual feedback effects for target acquisition, improving game affordance and interaction quality. Furthermore, we plan to implement context-sensitive HUD visibility to maximize immersion.

To highlight local biodiversity and enhance educational content, we plan to incorporate 3D models of Hong Kong's unique marine species, including Fiddler Crabs, Mudskippers, Green Sea Turtles, and Chinese White Dolphins as rescue targets. Additionally, we aim to recreate distinctive Hong Kong marine landscapes, such as Clear Water Bay and Hoi Ha Wan Marine Park, to provide an authentic local context for the game environment.

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REFERENCES

- [1] P. Boffi, M. Clerici, M. Muolo, A. Gallace, and P. L. Lanzi. Envisioning corals: Embodying coral reef inhabitants to raise awareness on climate changes impacts on remote environments. In *2023 9th International Conference on Virtual Reality (ICVR)*, pp. 239–246. IEEE, 2023. 1
- [2] J. Bosch, P. Ridao, D. Ribas, and N. Gracias. Creating 360 underwater virtual tours using an omnidirectional camera integrated in an auv. In *Oceans 2015-Genova*, pp. 1–7. IEEE, 2015. 1
- [3] L. Calvi, C. P. Santos, J. Relouw, B. Endrovski, C. Rothwell, A. Sarà, S. Lucrezi, M. Palma, and U. Pantaleo. A vr game to teach underwater sustainability while diving. In *2017 Sustainable Internet and ICT for Sustainability (SustainIT)*, pp. 1–4. IEEE, 2017. 1
- [4] V. Kövecses-Gösi. The pedagogical project of education for sustainable development in 3d virtual space. In *2019 10th IEEE International Conference on Cognitive Infocommunications (CogInfoCom)*, pp. 000539–000544. IEEE, 2019. 1
- [5] I. Lee. The metaverse: Ecosystem, technologies, business opportunities, and challenges. *IT Professional*, 26(5):28–35, 2024. 1
- [6] L. Nisiotis, A. Piki, P. Theodorou, S. Leal, L. C. Barradas, and J. A. Nascimento. Evaluation of a serious vr game designed to promote the sustainable development goals. In *2024 IEEE Gaming, Entertainment, and Media Conference (GEM)*, pp. 1–6. IEEE, 2024. 1